

Yash Rawal

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Professional Summary

Applied AI & Machine Learning Engineer serving a **1M+ user scale** by architecting enterprise agentic platforms, clinical NLP pipelines, and Model Context Protocol (MCP) tool gateways across 120+ organizations. Specialized in LLM orchestration (LangGraph, ReAct loops), healthcare data summarization (116K+ records), automated LLM-as-a-judge evaluation control planes, and PII sanitization. Active contributor to Spring AI.

Technical Skills

Languages & Backend: Python, Java, SQL, Spring Boot, FastAPI, PostgreSQL, pgvector, WebSockets, REST APIs

Agentic AI & Systems: Agentic Architectures, Multi-Agent Systems, ReAct Loops, Model Context Protocol (MCP), Tool RAG, A2A Protocol

Healthcare & NLP: Clinical Record Summarization, Healthcare NLP, FHIR Data Extraction, PII Masking (BIO NER), Prompt Evals

AI & ML Frameworks: LangChain, LangGraph, Spring AI, PyTorch, vLLM, LlamaIndex, GROBID, pdfplumber, Hugging Face

Platforms & Infrastructure: Docker, Kubernetes, Linux, CI/CD, LLM-as-Judge, Red Teaming, Azure OpenAI, Bedrock, Vertex AI, Claude, Llama, Qwen

Experience

Bigscal Technologies Pvt. Ltd.
AI/ML Engineer Oct 2024 - Present
Surat, India

- Deployed a production-scale agentic platform and multi-agent systems serving **1M+ active users** across 120+ enterprise tenants on cloud infrastructure (Azure, AWS, GCP).
- Architected multi-agent workflows with LangChain and LangGraph, utilizing **MCP Server Tool Routing (Tool RAG)** and A2A communication with integrated guardrails.
- Designed a capability contract layer to execute nested profiles inside a single ReAct loop, reducing execution overhead by **3x** and token consumption by 70%.
- Implemented a hierarchical RAG pipeline indexing 10,000+ technical documents using GROBID, pdfplumber, and PostgreSQL/PGVector, achieving a **90%+ Recall@10** retrieval rate.
- Engineered an automated LLM evaluation and red-teaming control plane to log reasoning trajectories, scoring agent behaviors via **LLM-as-a-judge** and RLHF preference datasets.
- Built production guardrails against prompt injections and data leaks, integrating runtime tool validation and **automated PII redaction**.

Bigscal Technologies Pvt. Ltd.
AI/ML Engineer Intern Apr 2024 - Sep 2024
Surat, India

- Engineered clinical summarization pipelines for **116K+ medical records** with 4-bit quantized Llama 3, integrating automated PII sanitization (BIO NER) for compliant healthcare data processing.
- Designed real-time WebSocket audio streaming for Indic ASR with **NeMo RNN-T and CTC decoding**, compiling custom Indic LJSpeech-format datasets.

Open Source & Projects

Modular Agent Harness & Memory System ("Brain") | Open Source
Principal Creator & Architect of an open-source plug-and-play agentic framework designed as modular building blocks for flexible workflows, implementing formal ReAct loops with step budgets, pgvector tool routing, and dynamic MCP settings caching.
Java, Spring Boot, PostgreSQL, WebSockets

Spring AI (spring-projects/spring-ai) | Open Source
Merged PR #5711 to propagate reasoningContent from AWS Bedrock and OpenAI-compatible models, and resolved Converse API prompt-caching TTL latency consistency.
Java, Spring Boot, Python

Mini Language Model (~85M Parameter Architecture) | Personal Research
Pretrained an ~85M-parameter transformer to ~1.43 validation loss on TinyStories targeting **edge hardware deployment**; engineered FragmentStream chunked attention to eliminate $O(T^2)$ VRAM overhead and custom Triton kernels with KV caching.
Python, PyTorch, CUDA, Triton, Tesla P100/T4

Education

Master of Computer Applications (Artificial Intelligence & Machine Learning) - Parul University 2022 - 2024

Certifications

Microsoft Certified: Azure AI Fundamentals (AI-900)	Microsoft
Microsoft Certified: Azure Fundamentals (AZ-900)	Microsoft